Crib Sheet

Lookup Table of Theorems and Properties

Uniting Theorems	
1	$XY + XY' = X$
1D	$(X + Y)(X + Y') = X$
Absorption Theorems	
2	$X + XY = X$
2D	$X(X + Y) = X$
Elimination Theorems	
3	$X + X'Y = X + Y$
3D	$X(X' + Y) = XY$
Duality	
4	$(X + Y + Z + \cdots)^D = X \cdot Y \cdot Z \cdot \ldots$
4D	$(X \cdot Y \cdot Z \cdot \ldots)^D = X + Y + Z + \cdots$
Theorems for Multiplying out and Factoring	
5	$(X + Y)(X' + Z) = XZ + X'Y$
5D	$XY + X'Z = (X + Z)(X' + Y)$
Consensus Theorems	
6	$XY + YZ + X'Z = XY + X'Z$
6D	$(X + Y)(Y + Z)(X' + Z) = (X + Y)(X' + Z)$
Operations with 0 and 1	
1	$X + 0 = X$
1D	$X \cdot 1 = X$
2	$X + 1 = 1$
2D	$X \cdot 0 = 0$
Idempotent Laws	
3	$X + X = X$
3D	$X \cdot X = X$
Involution Law	
4	$(X')' = X$
Laws of Complementarity	
5	$X + X' = 1$
5D	$X \cdot X' = 0$
Commutative Laws	
6	$X + Y = Y + X$
6D	$X \cdot Y = Y \cdot X$
Associative Laws	
7	$(X + Y) + Z = X + (Y + Z) = X + Y + Z$
7D	$(X \cdot Y) \cdot Z = X \cdot (Y \cdot Z) = X \cdot Y \cdot Z$
Distributive Laws	
8	$X(Y + Z) = XY + XZ$
8D	$X + YZ = (X + Y)(X + Z)$
DeMorgan's Laws	
9	$(X + Y)' = X'Y'$
9D	$(X \cdot Y)' = X' + Y'$
Inequivalence (!=) / XOR	
$(X \oplus Y) = X'Y + XY'$	
Equivalence (==) / XNOR	
$(X \equiv Y) = XY + X'Y'$	

Term Numbering on Karnaugh Maps

		AB			
		00	01	11	10
CD	00	m_0	m_4	m_{12}	m_8
	01	m_1	m_5	m_{13}	m_9
	11	m_3	m_7	m_{15}	m_{11}
	10	m_2	m_6	m_{14}	m_{10}

<u>Prime Implicant (PI)</u>: a product in a SOP that cannot have any more letters removed without invalidating the expression and is, therefore, as simplified as possible. In the QM table, these are rows that cannot be combined any further.

<u>Essential Prime Implicant (EPI)</u>: a product in a SOP that includes minterms not implied by any other PI and therefore cannot be removed without invalidating the function.

<u>Maximum Pull-Up and Pull-Down Resistances:</u>

$$R_{PU} \leq \frac{(\mathrm{V_{CC}} - \mathrm{V_{IH_{min}}})}{\mathrm{I_{IH_{max}}}} \quad , \quad R_{PD} \leq \frac{\mathrm{V_{IL_{max}}}}{\mathrm{I_{IL_{max}}}}$$